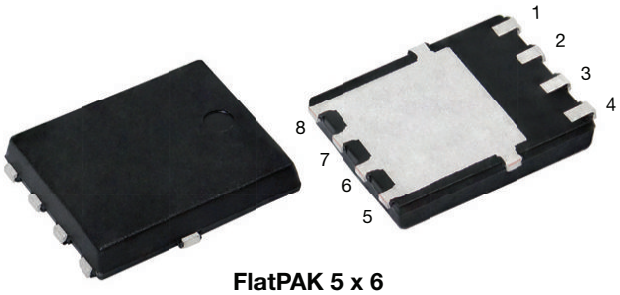
# High Current Density Surface-Mount Schottky Barrier Rectifier

Ultra Low $V_F$ = 0.52 V at $I_F$ = 5 A




**FlatPAK 5 x 6**




## FEATURES

- Guardring for overvoltage protection
- High barrier technology, $T_J$ = 175 °C maximum
- Low leakage current
- Enhanced for high surge endurance
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
  - Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE Available

RoHS COMPLIANT HALOGEN FREE

## LINKS TO ADDITIONAL RESOURCES

3D Models | Packages | Marking

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 20 A |
| $V_{RRM}$ | 100 V |
| $I_{FSM}$ | 400 A |
| $V_F$ at $I_F$ = 20 A ($T_J$ = 125 °C) | 0.65 V |
| $T_J$ max. | 175 °C |
| Package | FlatPAK 5 x 6 |
| Circuit configuration | Single |

## TYPICAL APPLICATIONS

- Freewheeling diode for hot-swap circuit
- Snubber diode for adsorbing excess energy
- AC-DC and DC-DC converters
- Boost and bulk diode for LED lighting
- Reverse polarity protection

## MECHANICAL DATA

**Case:** FlatPAK 5 x 6

Molding compound meets UL 94 V-0 flammability rating

Base P/N-M3 - halogen-free, RoHS-compliant

Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

**Terminals:** matte tin plated leads, solderable per J-STD-002

M3 and HM3 suffix meets JESD 201 class 2 whisker test

| **MAXIMUM RATINGS** ($T_A$ = 25 °C unless otherwise noted) | | | | |
|---|---|---|---|---|
| **PARAMETER** | | **SYMBOL** | **SS20KH102** | **UNIT** |
| Device marking code | | | 20H102 | |
| Maximum repetitive peak reverse voltage | | $V_{RRM}$ | 100 | V |
| Maximum average forward rectified current (fig.1) | | $I_{F(AV)}$ (1) | 20 | A |
| | | $I_{F(AV)}$ (2) | 3.7 | |
| Non-repetitive peak forward surge current | 8.3 ms half-sine wave superimposed on rated load | $I_{FSM}$ | 400 | |
| | 100 µs square pulse | | 1600 | |
| Operating junction temperature range | | $T_J$ (3) | -40 to +175 | °C |
| Storage temperature range | | $T_{STG}$ | -55 to +175 | |

**Notes**

(1) With infinite heatsink

(2) Free air, mounted on recommended copper pad area, 2 oz., FR4 PCB

(3) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$



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| ELECTRICAL CHARACTERISTICS ($T_J$ = 25 °C unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | TEST CONDITIONS | | SYMBOL | TYP. | MAX. | UNIT |
| Instantaneous forward voltage | $I_F$ = 5 A | $T_J$ = 25 °C | $V_F$ (1) | 0.66 | - | V |
| | $I_F$ = 10 A | | | 0.73 | - | |
| | $I_F$ = 20 A | | | 0.80 | 0.84 | |
| | $I_F$ = 5 A | $T_J$ = 125 °C | | 0.52 | - | |
| | $I_F$ = 10 A | | | 0.58 | - | |
| | $I_F$ = 20 A | | | 0.65 | 0.69 | |
| Reverse current | $V_R$ = 70 V | $T_J$ = 25 °C | $I_R$ (2) | 0.06 | - | µA |
| | | $T_J$ = 125 °C | | 280 | - | |
| | $V_R$ = 100 V | $T_J$ = 25 °C | | - | 6 | |
| | | $T_J$ = 125 °C | | 550 | 2000 | |
| Typical junction capacitance | 4.0 V, 1 MHz | | $C_J$ | 390 | - | pF |

**Notes**

(1) Pulse test: 300 µs pulse width, 1 % duty cycle
(2) Pulse test: pulse width ≤ 5 ms

| THERMAL CHARACTERISTICS ($T_A$ = 25 °C unless otherwise noted) | | | | |
|---|---|---|---|---|
| PARAMETER | SYMBOL | TYP. | MAX. | UNIT |
| Thermal resistance | $R_{\theta JA}$ (1)(2) | 75 | - | °C/W |
| | $R_{\theta JM}$ (3) | 2.5 | 3.5 | |

**Notes**

(1) The heat generated must be less than thermal conductivity from junction to ambient: $dP_D/dT_J < 1/R_{\theta JA}$
(2) Free air, mounted on recommended copper pad area; thermal resistance $R_{\theta JA}$ - junction-to-ambient
(3) Mounted on infinite heatsink; thermal resistance $R_{\theta JM}$ - junction-to-mount

## ORDERING INFORMATION TABLE

Device code

| SS | 20 | K | H | 10 | 2 | H | M3 |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

1 - Vishay planar Schottky product
2 - Current rating (20 = 20 A)
3 - Package type (K = FlatPAK 5 x 6)
4 - Process type option (H = low $I_R$)
5 - Voltage rating (10 = 100 V)
6 - Planar Schottky generation option (2 = Gen 2)
7 - Quality grade (H = AEC-Q101 qualified, otherwise = industry grade)
8 - Material / environmental category (M3 = halogen-free, RoHS-compliant, and termination lead (Pb)-free)

| ORDERING INFORMATION (Example) | | | | |
|---|---|---|---|---|
| PREFERRED P/N | UNIT WEIGHT (g) | PREFERRED PACKAGE CODE | BASE QUANTITY | DELIVERY MODE |
| SS20KH102-M3/H | 0.10 | H | 1500 | 7" diameter plastic tape and reel |
| SS20KH102-M3/I | 0.10 | I | 6000 | 13" diameter plastic tape and reel |
| SS20KH102-M3/IE (2) | 0.10 | IE | 6000 | 13" diameter plastic tape and reel |
| SS20KH102HM3/H (1) | 0.10 | H | 1500 | 7" diameter plastic tape and reel |
| SS20KH102HM3/I (1) | 0.10 | I | 6000 | 13" diameter plastic tape and reel |
| SS20KH102HM3/IE (1)(2) | 0.10 | IE | 6000 | 13" diameter plastic tape and reel |

**Note**

(1) AEC-Q101 qualified
(2) Component rotate 90° in cavity


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## RATINGS AND CHARACTERISTICS CURVES ($T_A$ = 25 °C unless otherwise noted)

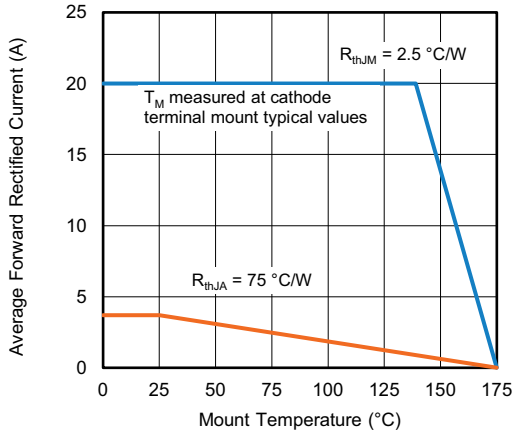



Fig. 1 - Maximum Forward Current Derating Curve

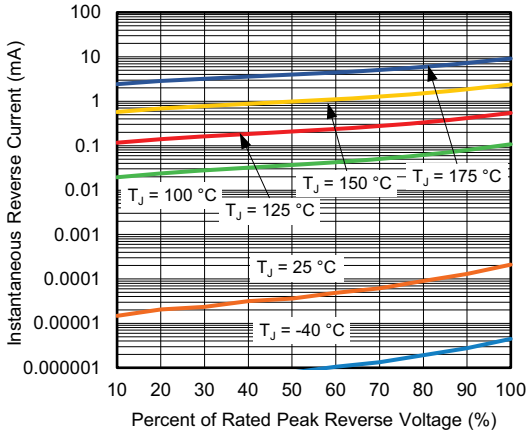



Fig. 4 - Typical Reverse Leakage Characteristics

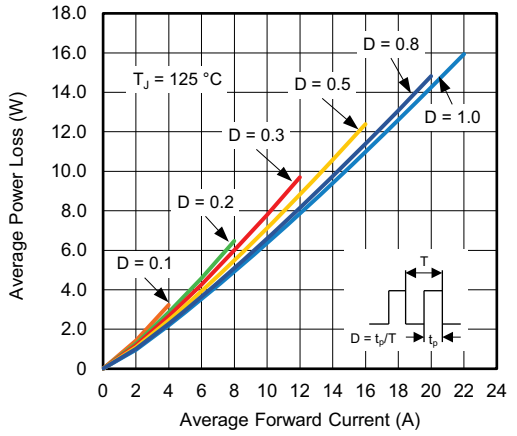



Fig. 2 - Forward Power Loss Characteristics

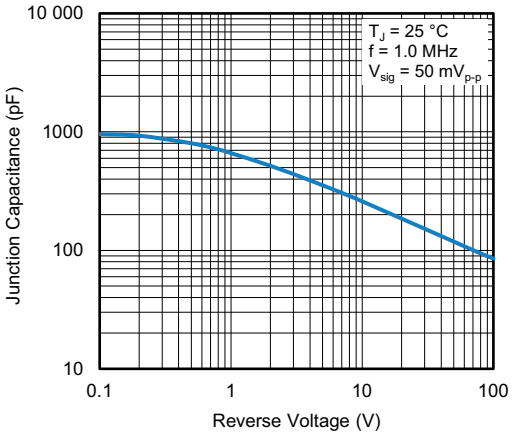



Fig. 5 - Typical Junction Capacitance

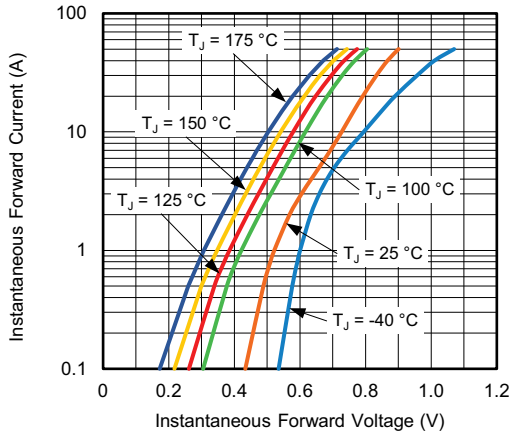



Fig. 3 - Typical Instantaneous Forward Characteristics

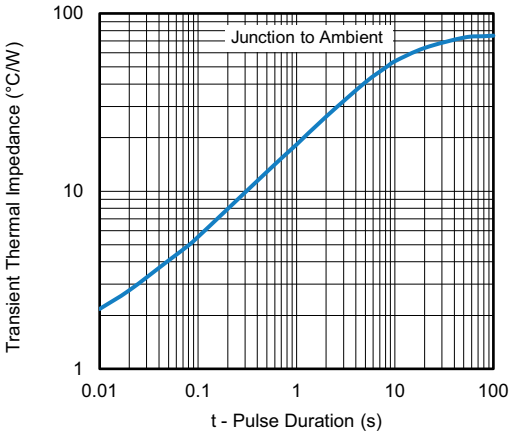
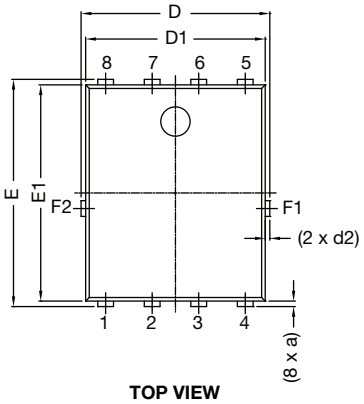



Fig. 6 - Typical Transient Thermal Impedance

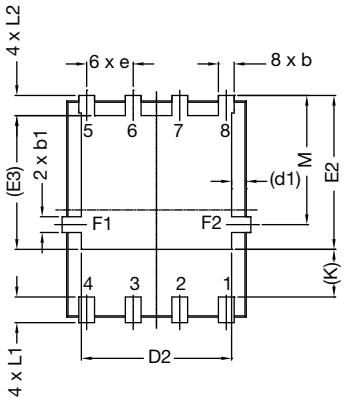

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**DIMENSIONS** in inches (millimeters)

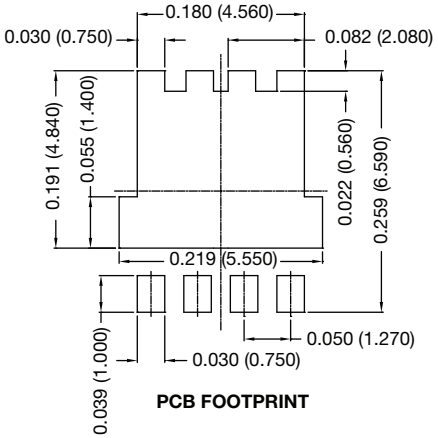
## FlatPAK 5 x 6



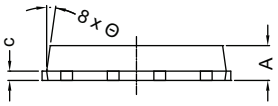

TOP VIEW




BOTTOM VIEW




PCB FOOTPRINT




SIDE VIEW

| DIM. | INCHES | | | MILLIMETERS | | |
|---|---|---|---|---|---|---|
| | MIN. | NOM. | MAX. | MIN. | NOM. | MAX. |
| A | 0.035 | 0.039 | 0.043 | 0.89 | 0.99 | 1.09 |
| (a) | - | 0.006 | - | - | 0.15 | - |
| b | 0.013 | 0.017 | 0.020 | 0.32 | 0.43 | 0.52 |
| b1 | 0.013 | 0.017 | 0.020 | 0.32 | 0.43 | 0.52 |
| c | 0.008 | - | 0.014 | 0.20 | - | 0.35 |
| D | 0.197 | 0.203 | 0.209 | 5.00 | 5.15 | 5.30 |
| D1 | 0.189 | 0.193 | 0.197 | 4.80 | 4.90 | 5.00 |
| D2 | 0.154 | 0.161 | 0.169 | 3.90 | 4.10 | 4.30 |
| (d1) | - | 0.016 | - | - | 0.40 | - |
| (d2) | - | 0.005 | - | - | 0.125 | - |
| E | 0.238 | 0.244 | 0.250 | 6.05 | 6.20 | 6.35 |
| E1 | 0.228 | 0.232 | 0.236 | 5.80 | 5.90 | 6.00 |
| E2 | 0.157 | 0.165 | 0.173 | 4.00 | 4.20 | 4.40 |
| (E3) | - | 0.144 | - | - | 3.65 | - |
| e | 0.050 BSC | | | 1.27 BSC | | |
| (K) | 0.039 | - | - | 1.00 | - | - |
| L1 | 0.019 | - | 0.043 | 0.48 | - | 1.10 |
| L2 | 0.012 | - | 0.031 | 0.30 | - | 0.80 |
| M | 0.128 | 0.138 | 0.148 | 3.25 | 3.50 | 3.75 |
| Θ | 0° | - | 10° | 0° | - | 10° |

**Notes**

- Dimensioning and tolerancing per ASME Y14.5-2009
- Dimensions D1 and E1 do not include mold flash or gate burrs
- Dimension (XX) means reference only



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